

Natural Language Processing

Lecture 08: Large Language Models

Valentin Malykh, MTS AI



Autumn 2024

The course is delivered at ITMO University, Saint-Petersburg,
Bauman University, Moscow, IITU, Almaty

Content

- 1 Pretrained Language Models
- 2 Large Language Models
- 3 Parameter-Efficient Fine-Tuning

Content

- 1 Pretrained Language Models
- 2 Large Language Models
- 3 Parameter-Efficient Fine-Tuning

Content

1 Pretrained Language Models

- General introduction
- Encoders Recap
- Encoder-Decoder PLMs
- Decoder PLMs

How can we define a language? (Recap)

— The probabilistic approach

- A language can also be defined as a probabilistic distribution over all the possible sentences.
- A statistical language model is a probability distribution over sequences of words (sentences) in a given language L :

$$\sum_{s \in V^+} P_{LM}(s) = 1$$

- Or:

$$\sum_{\substack{s=w_1 w_2 \dots w_n \\ w_i \in V, n > 0}} P_{LM}(s) = 1$$

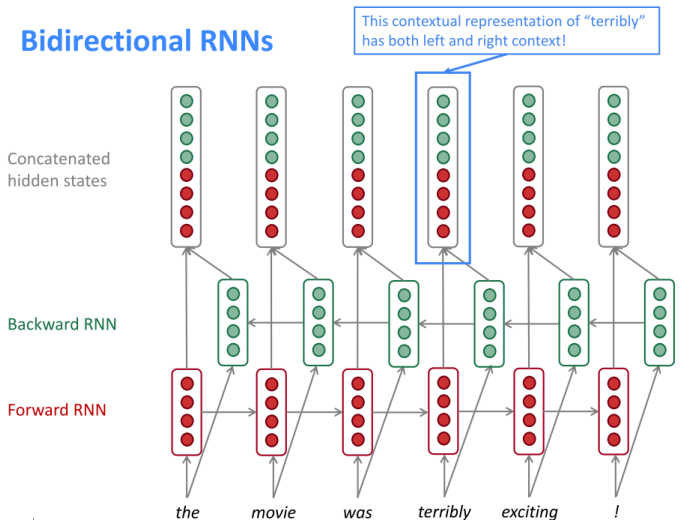
NLP tasks as Conditional LMs (Recap)

x “input”	w “text output”
An author	A document written by that author
A topic label	An article about that topic
{SPAM, NOT_SPAM}	An email
A sentence in French	Its English translation
A sentence in English	Its French translation
A sentence in English	Its Chinese translation
An image	A text description of the image
A document	Its summary
A document	Its translation
Meteorological measurements	A weather report
Acoustic signal	Transcription of speech
Conversational history + database	Dialogue system response
A question + a document	Its answer
A question + an image	Its answer

Chris Dyer, Conditional LMs (slides)

Bidirectional RNN Language Models (Recap)

Bidirectional RNNs

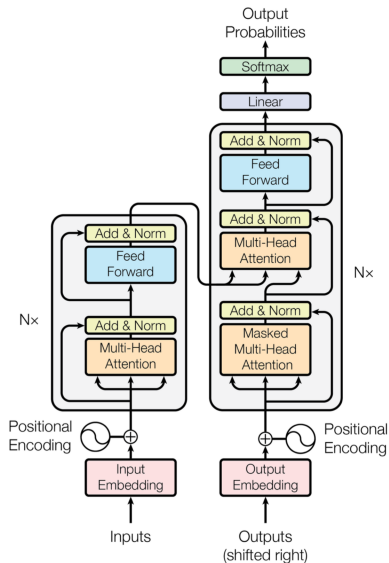


Christopher Manning, Natural Language Processing with Deep Learning, Stanford U. CS224n

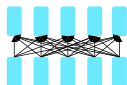
Transformer Recap

- Transformer NMT adopts a sequence-to-sequence architecture.
- The figure from Jay Alammar's blog.

<http://jalamar.github.io/illustrated-transformer/>

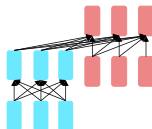


The neural architecture influences the type of pretraining, and natural use cases.



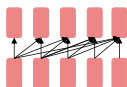
Encoders

- Gets bidirectional context – can condition on future!
- How do we train them to build strong representations?



Encoder- Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?



Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words

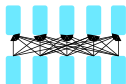
Tatsunori Hashimoto, Natural Language Processing (slides)

Content

1 Pretrained Language Models

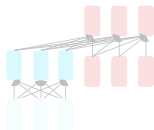
- General introduction
- **Encoders Recap**
- Encoder-Decoder PLMs
- Decoder PLMs

The neural architecture influences the type of pretraining, and natural use cases.



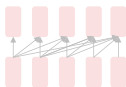
Encoders

- Gets bidirectional context – can condition on future!
- How do we train them to build strong representations?



Encoder- Decoders

- Good parts of decoders and encoders?
- What's the best way to pretrain them?



Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words

Tatsunori Hashimoto, Natural Language Processing (slides)

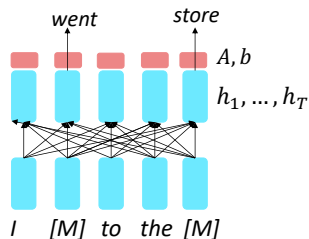
So far, we've looked at language model pretraining. But **encoders get bidirectional context**, so we can't do language modeling!

Idea: replace some fraction of words in the input with a special [MASK] token; predict these words.

$$h_1, \dots, h_T = \text{Encoder}(w_1, \dots, w_T)$$

$$y_i \sim Aw_i + b$$

Only add loss terms from words that are “masked out.” If $\tilde{x}$ is the masked version of x , we're learning $p_\theta(x|\tilde{x})$. Called **Masked LM**.

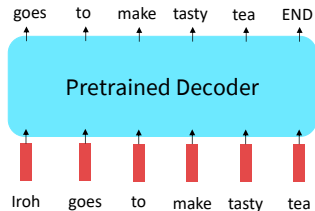
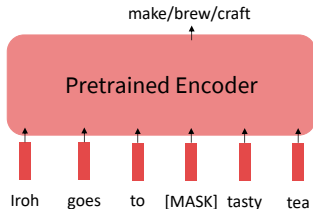


[Devlin et al., 2018]

Tatsunori Hashimoto, Natural Language Processing (slides)

Those results looked great! Why not use pretrained encoders for everything?

If your task involves generating sequences, consider using a pretrained decoder; BERT and other pretrained encoders don't naturally lead to nice autoregressive (1-word-at-a-time) generation methods.



Tatsunori Hashimoto, Natural Language Processing (slides)

Content

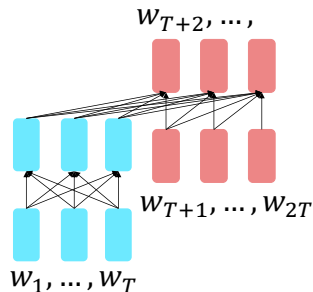
1 Pretrained Language Models

- General introduction
- Encoders Recap
- **Encoder-Decoder PLMs**
- Decoder PLMs

For **encoder-decoders**, we could do something like **language modeling**, but where a prefix of every input is provided to the encoder and is not predicted.

$$\begin{aligned}
 h_1, \dots, h_T &= \text{Encoder}(w_1, \dots, w_T) \\
 h_{T+1}, \dots, h_{2T} &= \text{Decoder}(w_1, \dots, w_T, h_1, \dots, h_T) \\
 y_i &\sim A h_i + b, i > T
 \end{aligned}$$

The **encoder** portion benefits from bidirectional context; the **decoder** portion is used to train the whole model through language modeling.



[Raffel et al., 2018]

Tatsunori Hashimoto, Natural Language Processing (slides)

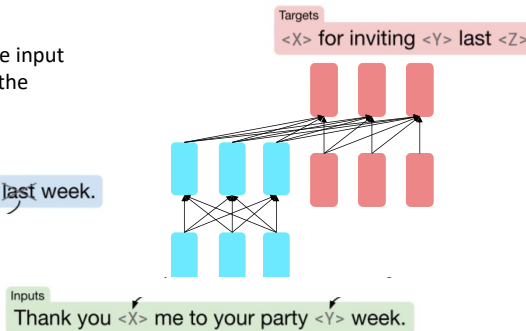
What [Raffel et al., 2018](#) found to work best was **span corruption**. Their model: **T5**.

Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!

Original text

Thank you for inviting me to your party last week.

This is implemented in text preprocessing: it's still an objective that looks like **language modeling** at the decoder side.



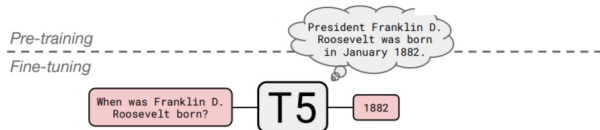
Tatsunori Hashimoto, Natural Language Processing (slides)

[Raffel et al., 2018](#) found encoder-decoders to work better than decoders for their tasks, and span corruption (denoising) to work better than language modeling.

Architecture	Objective	Params	Cost	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Encoder-decoder	Denoising	$2P$	M	83.28	19.24	80.88	71.36	26.98	39.82	27.65
Enc-dec, shared	Denoising	P	M	82.81	18.78	80.63	70.73	26.72	39.03	27.46
Enc-dec, 6 layers	Denoising	P	$M/2$	80.88	18.97	77.59	68.42	26.38	38.40	26.95
Language model	Denoising	P	M	74.70	17.93	61.14	55.02	25.09	35.28	25.86
Prefix LM	Denoising	P	M	81.82	18.61	78.94	68.11	26.43	37.98	27.39
Encoder-decoder	LM	$2P$	M	79.56	18.59	76.02	64.29	26.27	39.17	26.86
Enc-dec, shared	LM	P	M	79.60	18.13	76.35	63.50	26.62	39.17	27.05
Enc-dec, 6 layers	LM	P	$M/2$	78.67	18.26	75.32	64.06	26.13	38.42	26.89
Language model	LM	P	M	73.78	17.54	53.81	56.51	25.23	34.31	25.38
Prefix LM	LM	P	M	79.68	17.84	76.87	64.86	26.28	37.51	26.76

Tatsunori Hashimoto, Natural Language Processing (slides)

A fascinating property of T5: it can be finetuned to answer a wide range of questions, retrieving knowledge from its parameters.



		NQ	WQ	TQA		
				dev	test	
NQ: Natural Questions	Karpukhin et al. (2020)	41.5	42.4	57.9	–	
WQ: WebQuestions	T5.1.1-Base	25.7	28.2	24.2	30.6	220 million params
	T5.1.1-Large	27.3	29.5	28.5	37.2	770 million params
TQA: Trivia QA	T5.1.1-XL	29.5	32.4	36.0	45.1	3 billion params
	T5.1.1-XXL	32.8	35.6	42.9	52.5	11 billion params
All “open-domain”	T5.1.1-XXL + SSM	35.2	42.8	51.9	61.6	

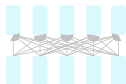
Tatsunori Hashimoto, Natural Language Processing (slides)

Content

1 Pretrained Language Models

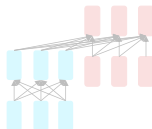
- General introduction
- Encoders Recap
- Encoder-Decoder PLMs
- Decoder PLMs

The neural architecture influences the type of pretraining, and natural use cases.



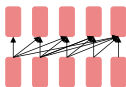
Encoders

- Gets bidirectional context – can condition on future!
- How do we train them to build strong representations?



**Encoder-
Decoders**

- Good parts of decoders and encoders?
- What's the best way to pretrain them?



Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words
- All the biggest pretrained models are Decoders.

Tatsunori Hashimoto, Natural Language Processing (slides)

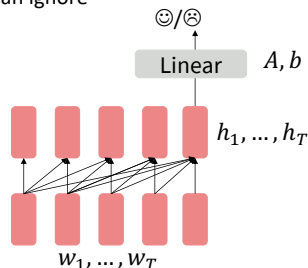
When using language model pretrained decoders, we can ignore that they were trained to model $p(w_t|w_{1:t-1})$.

We can finetune them by training a classifier on the last word's hidden state.

$$h_1, \dots, h_T = \text{Decoder}(w_1, \dots, w_T)$$
$$y \sim Ah_T + b$$

Where A and b are randomly initialized and specified by the downstream task.

Gradients backpropagate through the whole network.



Tatsunori Hashimoto, Natural Language Processing (slides)

It's natural to pretrain decoders as language models and then use them as generators, finetuning their $p_{\theta}(w_t|w_{1:t-1})!$

This is helpful in tasks **where the output is a sequence** with a vocabulary like that at pretraining time!

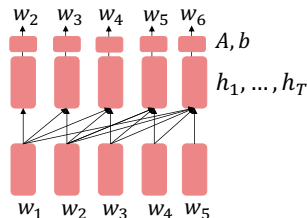
- Dialogue (context=dialogue history)
- Summarization (context=document)

$$h_1, \dots, h_T = \text{Decoder}(w_1, \dots, w_T)$$

$$w_t \sim Ah_{t-1} + b$$

Where A, b were pretrained in the language model!

Tatsunori Hashimoto, Natural Language Processing (slides)



[Note how the linear layer has been pretrained.]

GPT: introduction

- Following the similar idea of ELMo, OpenAI GPT, short for Generative Pre-training Transformer (Radford et al., 2018), expands the unsupervised language model to a much larger scale by training on a giant collection of free text corpora.
- Despite of the similarity, GPT has two major differences from ELMo:
 - The model architectures are different: ELMo uses a shallow concatenation of independently trained left-to-right and right-to-left multi-layer LSTMs, while GPT is a multi-layer transformer decoder.
 - The use of contextualized embeddings in downstream tasks are different: ELMo feeds embeddings into models customized for specific tasks as additional features, while GPT fine-tunes the same base model for all end tasks.

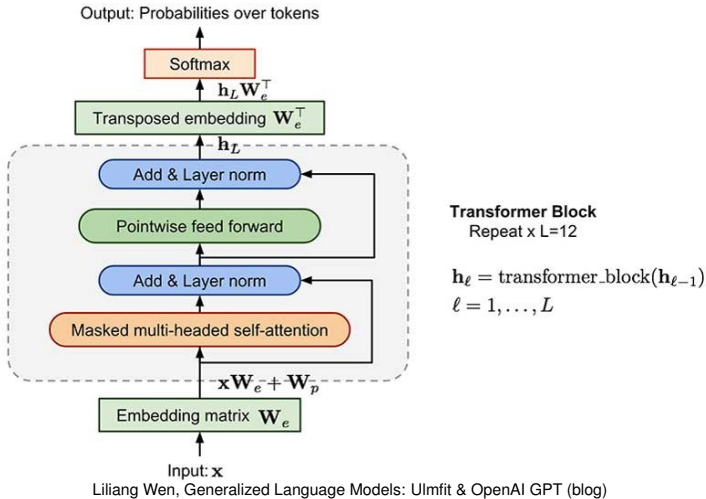
Liliang Wen, Generalized Language Models: ULMFIT & OpenAI GPT (blog)

GPT: Transformer Decoder as Language Model

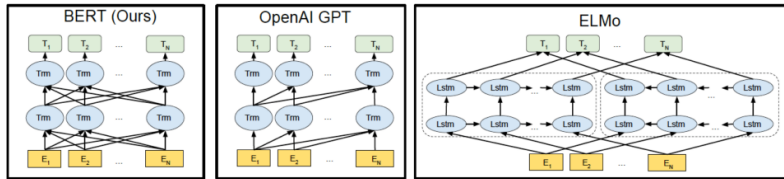
- Compared to the original transformer architecture, the transformer decoder model discards the encoder part, so there is only one single input sentence rather than two separate source and target sequences.
- This model applies multiple transformer blocks over the embeddings of input sequences. Each block contains a masked multi-headed self-attention layer and a pointwise feed-forward layer. The final output produces a distribution over target tokens after softmax normalization.

Liliang Wen, Generalized Language Models: Ulmfit & OpenAI GPT (blog)

GPT: Transformer Decoder as Language Model



Differences: BERT, GPT & ELMo



Differences in pre-training model architectures: BERT, OpenAI GPT, and ELMo

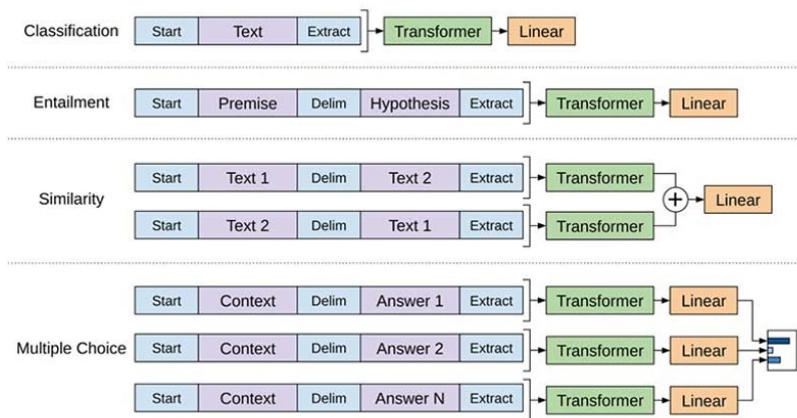
Devlin et al., 2018, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Slides)

GPT: BPE for subword segmentation

Byte Pair Encoding (BPE) is used to encode the input sequences. BPE was originally proposed as a data compression algorithm in 1990s and then was adopted to solve the open-vocabulary issue in machine translation, as we can easily run into rare and unknown words when translating into a new language. Motivated by the intuition that rare and unknown words can often be decomposed into multiple subwords, BPE finds the best word segmentation by iteratively and greedily merging frequent pairs of characters.

Liliang Wen, Generalized Language Models: Ulmfit & OpenAI GPT (blog)

GPT: supervised fine-tuning



Liliang Wen, Generalized Language Models: Ulmfit & OpenAI GPT (blog)

GPT-2: Introduction

- The OpenAI GPT-2 language model is a direct successor to GPT. GPT-2 has 1.5B parameters, 10x more than the original GPT, and it achieves SOTA results on 7 out of 8 tested language modeling datasets in a zero-shot transfer setting without any task-specific fine-tuning.
- The pre-training dataset contains 8 million Web pages collected by crawling qualified outbound links from Reddit. Large improvements by OpenAI GPT-2 are specially noticeable on small datasets and datasets used for measuring long-term dependency.

Liliang Wen, Generalized Language Models: Bert & OpenAI GPT-2 (blog)

GPT-2: BPE on Byte Sequence

- Same as the original GPT, GPT-2 uses BPE but on UTF-8 byte sequences.
- BPE merges frequently co-occurred byte pairs in a greedy manner. To prevent it from generating multiple versions of common words (i.e. dog., dog! and dog? for the word dog), GPT-2 prevents BPE from merging characters across categories (thus dog would not be merged with punctuations like ., ! and ?). This tricks help increase the quality of the final byte segmentation.
- Using the byte sequence representation, GPT-2 is able to assign a probability to any Unicode string, regardless of any pre-processing steps.

Liliang Wen, Generalized Language Models: Bert & OpenAI GPT-2 (blog)

GPT-2: Model Modifications

Compared to GPT, other than having many more transformer layers and parameters, GPT-2 incorporates only a few architecture modifications:

- Layer normalization was moved to the input of each sub-block, similar to a residual unit of type “building block” (differently from the original type “bottleneck”, it has batch normalization applied before weight layers).
- An additional layer normalization was added after the final self-attention block.

Liliang Wen, Generalized Language Models: Bert & OpenAI GPT-2 (blog)

GPT-2: Model Modifications

Compared to GPT, other than having many more transformer layers and parameters, GPT-2 incorporates only a few architecture modifications:

- A modified initialization was constructed as a function of the model depth.
- The weights of residual layers were initially scaled by a factor of $\frac{1}{\sqrt{N}}$ where N is the number of residual layers.
- Use larger vocabulary size and context size.

Liliang Wen, Generalized Language Models: Bert & OpenAI GPT-2 (blog)

GPT-2: Models

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Table 2. Architecture hyperparameters for the 4 model sizes.

Liliang Wen, Generalized Language Models: Bert & OpenAI GPT-2 (blog)

GPT-2: Results

Language Models are Unsupervised Multitask Learners

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	21.8
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	1.17	37.50	75.20
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	1.06	26.37	55.72
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	1.02	22.05	44.575
1542M	8.63	63.24	93.30	89.05	18.34	35.76	0.93	0.98	17.48	42.16

Table 3. Zero-shot results on many datasets. No training or fine-tuning was performed for any of these results. PTB and WikiText-2 results are from (Gong et al., 2018). CBT results are from (Bajgar et al., 2016). LAMBADA accuracy result is from (Hoang et al., 2018) and LAMBADA perplexity result is from (Grave et al., 2016). Other results are from (Dai et al., 2019).

Alec Radford, et al., Language Models are Unsupervised Multitask Learners, OpenAI Blog 1.8 (2019): 9

So far, we've interacted with pretrained models in two ways:

- Sample from the distributions they define (maybe providing a prompt)
- Fine-tune them on a task we care about, and take their predictions.

Very large language models seem to perform some kind of learning **without gradient steps** simply from examples you provide within their contexts.

GPT-3 is the canonical example of this. The largest T5 model had 11 billion parameters.

GPT-3 has 175 billion parameters.

Tatsunori Hashimoto, Natural Language Processing (slides)

Very large language models seem to perform some kind of learning **without gradient steps** simply from examples you provide within their contexts.

The in-context examples seem to specify the task to be performed, and the conditional distribution mocks performing the task to a certain extent.

Input (prefix within a single Transformer decoder context):

“ thanks -> merci
 hello -> bonjour
 mint -> menthe
 otter -> ”

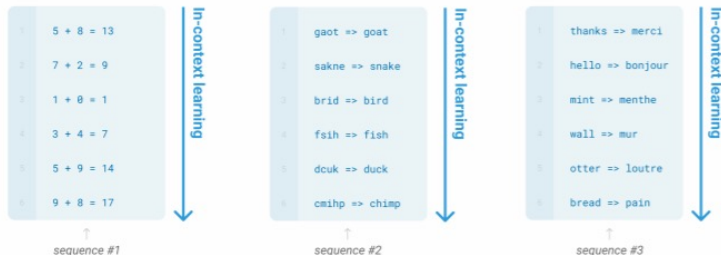
Output (conditional generations):

loutre...”

Tatsunori Hashimoto, Natural Language Processing (slides)

Very large language models seem to perform some kind of learning **without gradient steps** simply from examples you provide within their contexts.

Learning via SGD during unsupervised pre-training



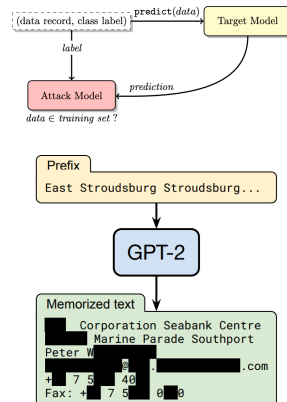
Tatsunori Hashimoto, Natural Language Processing (slides)

There's increasing evidence that pretrained models learn a wide variety of things about the statistical properties of language. Taking our examples from the start of class:

- *Stanford University is located in _____, California.* [Trivia]
- *I put ____ fork down on the table.* [syntax]
- *The woman walked across the street, checking for traffic over ____ shoulder.* [coreference]
- *I went to the ocean to see the fish, turtles, seals, and ____.* [lexical semantics/topic]
- *Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was ____.* [sentiment]
- *Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the ____.* [some reasoning – this is harder]
- *I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, ____* [some basic arithmetic; they don't learn the Fibonacci sequence]
- Models also learn – and can exacerbate racism, sexism, all manner of bad biases.

Tatsunori Hashimoto, Natural Language Processing (slides)

- *Membership inference* lets you recover parts of the training data
- Sometimes this training data is semi-private material from the web (addresses, emails)



Tatsunori Hashimoto, Natural Language Processing (slides)

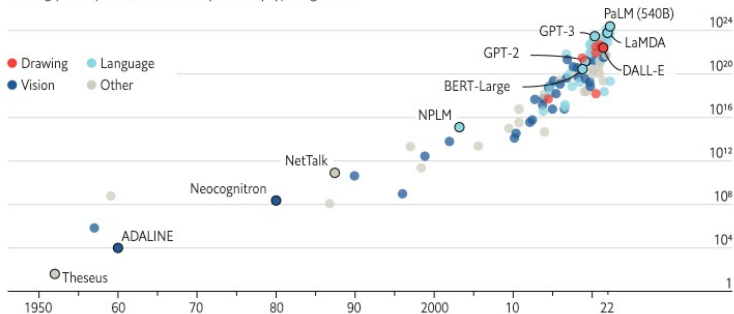
Content

- 1 Pretrained Language Models
- 2 Large Language Models**
- 3 Parameter-Efficient Fine-Tuning

The blessings of scale

AI training runs, estimated computing resources used

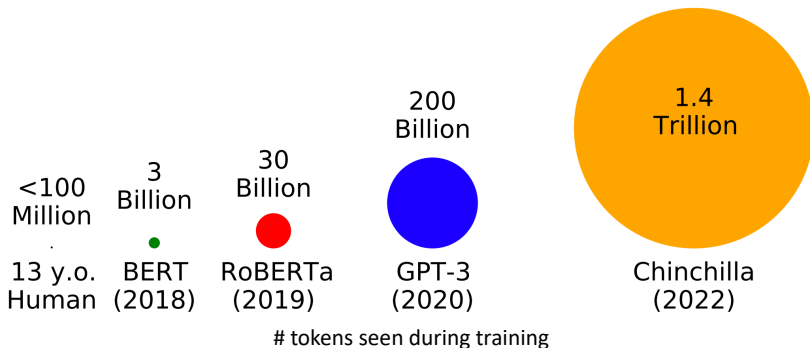
Floating-point operations, selected systems, by type, log scale



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

<https://www.economist.com/interactive/briefing/2022/06/11/huge-foundation-models-are-turbo-charging-ai-progress>

Tatsunori Hashimoto, Natural Language Processing (slides)



<https://babylm.github.io/>

Tatsunori Hashimoto, Natural Language Processing (slides)

Content

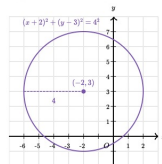
2 Large Language Models

- Emergent Abilities?
- Instruction Tuning
- Alignment

...*math*:

We can describe circles in the xy -plane using equations in terms of x and y . Circle equations questions require us to understand the connection between these equations and the features of circles.

For example, the equation $(x + 2)^2 + (y - 3)^2 = 4^2$ is graphed in the xy -plane below. It is a circle with a center at $(-2, 3)$ and a radius of 4.



<https://www.khanacademy.org/test-prep/sat/x0a8c2e5f:untitled-652>

Tatsunori Hashimoto, Natural Language Processing (slides)

...code:

```
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

<https://github.com/features/copilot>

Tatsunori Hashimoto, Natural Language Processing (slides)

...*medicine*:

Rapid and chronic ethanol tolerance are composed of distinct memory-like states in *Drosophila*

Abstract

Ethanol tolerance is the first type of behavioral plasticity and neural plasticity that is induced by ethanol intake, and yet its molecular and circuit bases remain largely unexplored. Here, we characterize three distinct forms of ethanol tolerance in male *Drosophila*: rapid, chronic, and repeated. Rapid tolerance is composed of two short-lived memory-like states, one that is labile and one that is consolidated. Chronic tolerance, induced by continuous exposure, lasts for two days, induces ethanol preference, and hinders the development of rapid tolerance through the activity of

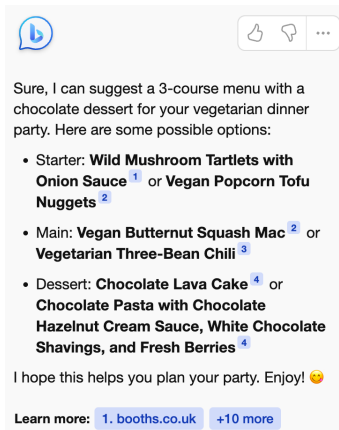
[[Larnerd, 2023](#)]

Tatsunori Hashimoto, Natural Language Processing (slides)

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

[[Microsoft Bing](#)]

(Also see OpenAI's ChatGPT, Google's Bard, Anthropic's Claude)



The screenshot shows a chat interface with a blue speech bubble icon on the left and thumbs up/down and a menu icon on the right. The text of the chat is as follows:

Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:

- Starter: **Wild Mushroom Tartlets with Onion Sauce** ¹ or **Vegan Popcorn Tofu Nuggets** ²
- Main: **Vegan Butternut Squash Mac** ² or **Vegetarian Three-Bean Chili** ³
- Dessert: **Chocolate Lava Cake** ⁴ or **Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries** ⁴

I hope this helps you plan your party. Enjoy! 😊

Learn more: [1. booths.co.uk](#) [+10 more](#)

Tatsunori Hashimoto, Natural Language Processing (slides)

Content

2 Large Language Models

- Emergent Abilities?
- **Instruction Tuning**
- Alignment

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].

Tatsunori Hashimoto, Natural Language Processing (slides)

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

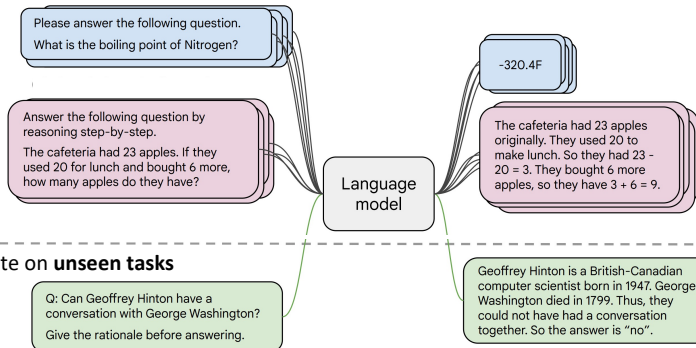
COMPLETION **Human**

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].
Finetuning to the rescue!

Tatsunori Hashimoto, Natural Language Processing (slides)

- **Collect examples** of (instruction, output) pairs across many tasks and finetune an LM



- **Evaluate on unseen tasks**

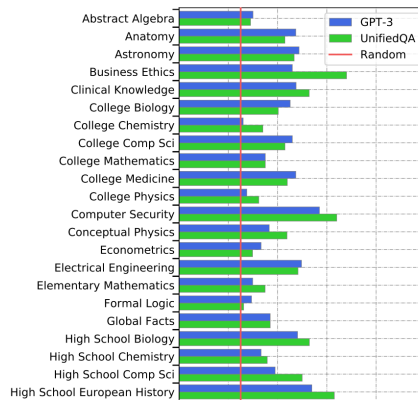
[FLAN-T5; [Chung et al., 2022](#)]

Tatsunori Hashimoto, Natural Language Processing (slides)

Massive Multitask Language Understanding (MMLU)

[[Hendrycks et al., 2021](#)]

New benchmarks for measuring LM performance on 57 diverse *knowledge intensive* tasks



Tatsunori Hashimoto, Natural Language Processing (slides)

Astronomy

What is true for a type-Ia supernova?

- A. This type occurs in binary systems.
- B. This type occurs in young galaxies.
- C. This type produces gamma-ray bursts.
- D. This type produces high amounts of X-rays.

Answer: A

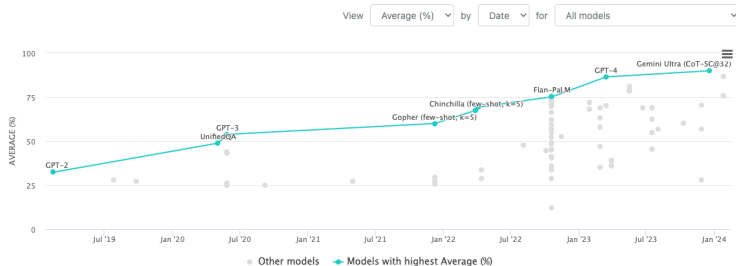
High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

- A. directional selection.
- B. stabilizing selection.
- C. sexual selection.
- D. disruptive selection

Answer: A

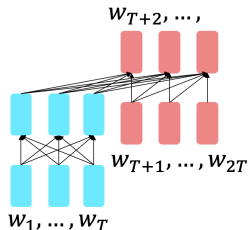
Tatsunori Hashimoto, Natural Language Processing (slides)



Rapid, impressive progress on challenging knowledge-intensive benchmarks

Tatsunori Hashimoto, Natural Language Processing (slides)

- Recall the T5 encoder-decoder model from lecture 10 [Raffel et al., 2018], pretrained on the **span corruption** task
- Flan-T5** [Chung et al., 2020]: T5 models finetuned on 1.8K additional tasks



		perplexity
80M	T5-Small	-9.2
	Flan-T5-Small	-3.1 (+6.1)
250M	T5-Base	-5.1
	Flan-T5-Base	6.5 (+11.6)
780M	T5-Large	-5.0
	Flan-T5-Large	13.8 (+18.8)
3B	T5-XL	-4.1
	Flan-T5-XL	19.1 (+23.2)
11B	T5-XXL	-2.9
	Flan-T5-XXL	23.7 (+26.6)

Bigger model = bigger Δ [\[Chung et al., 2022\]](#)

Tatsunori Hashimoto, Natural Language Processing (slides)

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.

The reporter and the chef will discuss the reporter's favorite dishes.

The reporter and the chef will discuss the chef's favorite dishes.

The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

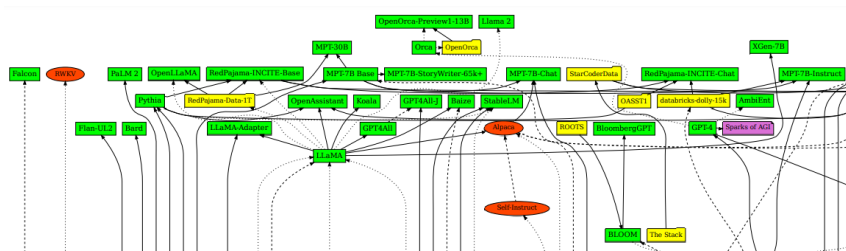
✖ (doesn't answer question)

Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl>

[Chung et al., 2022]

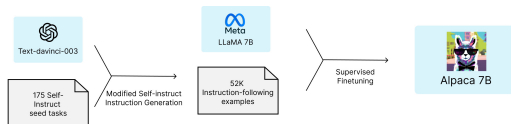
Tatsunori Hashimoto, Natural Language Processing (slides)



- The release of LLaMA led to open-source attempts to 'create' instruction tuning data

Tatsunori Hashimoto, Natural Language Processing (slides)

- You can generate data synthetically (from bigger LMs)



LIMA: Less Is More for Alignment

- You don't need many samples to instruction tune
- Crowdsourcing can be pretty effective!

Chunting Zhou^{1*} Pengfei Liu^{2*} Puxin Xu^{1†} Srinu Iyer^{1†} Jiao Sun²

Open Assistant

We believe we can create a revolution.

In the same way that Stable Diffusion helped the world make art and



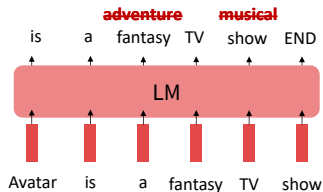
Tatsunori Hashimoto, Natural Language Processing (slides)

Content

2 Large Language Models

- Emergent Abilities?
- Instruction Tuning
- Alignment

- One limitation of instruction finetuning is obvious: it's **expensive** to collect ground-truth data for tasks.
- But there are other, subtler limitations too. Can you think of any?
- **Problem 1:** tasks like open-ended creative generation have no right answer.
 - *Write me a story about a dog and her pet grasshopper.*
- **Problem 2:** language modeling penalizes all token-level mistakes equally, but some errors are worse than others.
- Even with instruction finetuning, there a mismatch between the LM objective and the objective of "satisfy human preferences"!
- Can we **explicitly attempt to satisfy human preferences**?



Tatsunori Hashimoto, Natural Language Processing (slides)

- Let's say we were training a language model on some task (e.g. summarization).
- For each LM sample s , imagine we had a way to obtain a *human reward* of that summary: $R(s) \in \mathbb{R}$, higher is better.

SAN FRANCISCO,
California (CNN) --
A magnitude 4.2
earthquake shook the
San Francisco
...
overturn unstable
objects.

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$s_1 \\ R(s_1) = 8.0$$

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$s_2 \\ R(s_2) = 1.2$$

- Now we want to maximize the expected reward of samples from our LM:

$$\mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})]$$

Note: for mathematical simplicity
we're assuming only one "prompt"

Tatsunori Hashimoto, Natural Language Processing (slides)

Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

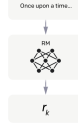
A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.

First step: instruction tuning!

Second + third steps: maximize reward (but how??)

Tatsunori Hashimoto, Natural Language Processing (slides)

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [[Phelps et al., 2015](#); [Clark et al., 2018](#)]

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

$$s_3$$
$$R(s_3) = 4.1? \quad 6.6? \quad 3.2?$$

Tatsunori Hashimoto, Natural Language Processing (slides)

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [Phelps et al., 2015; Clark et al., 2018]

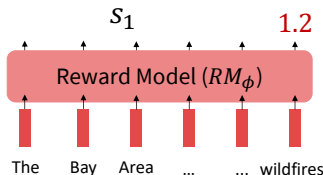
An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

>

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

>

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

 S_3 S_2

Bradley-Terry [1952] paired comparison model

$$J_{RM}(\phi) = -\mathbb{E}_{(s^w, s^l) \sim D} [\log \sigma(RM_\phi(s^w) - RM_\phi(s^l))]$$

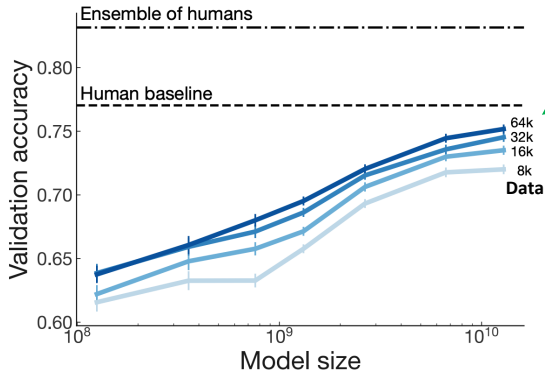
“winning”
sample

“losing”
sample

s^w should score
higher than s^l

Tatsunori Hashimoto, Natural Language Processing (slides)

Evaluate RM on predicting outcome of held-out human judgments



Large enough RM
trained on enough
data approaching
single human perf

[Stiennon et al., 2020]

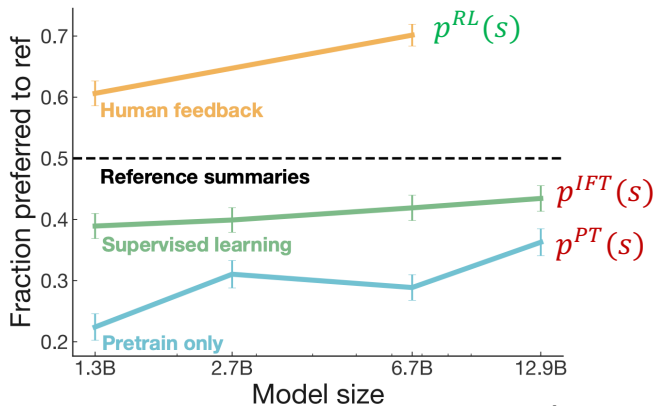
Tatsunori Hashimoto, Natural Language Processing (slides)

- Finally, we have everything we need:
 - A pretrained (possibly instruction-finetuned) LM $p^{PT}(s)$
 - A reward model $RM_\phi(s)$ that produces scalar rewards for LM outputs, trained on a dataset of human comparisons
 - A method for optimizing LM parameters towards an arbitrary reward function.
- Now to do RLHF:
 - Initialize a copy of the model $p_\theta^{RL}(s)$, with parameters θ we would like to optimize
 - Optimize the following reward with RL:

$$R(s) = RM_\phi(s) - \underbrace{\beta \log \left(\frac{p_\theta^{RL}(s)}{p^{PT}(s)} \right)}_{\text{Pay a price when } p_\theta^{RL}(s) > p^{PT}(s)}$$

This is a penalty which prevents us from diverging too far from the pretrained model. In expectation, it is known as the **Kullback-Leibler (KL) divergence** between $p_\theta^{RL}(s)$ and $p^{PT}(s)$.

Tatsunori Hashimoto, Natural Language Processing (slides)



[Stiennon et al., 2020]

Tatsunori Hashimoto, Natural Language Processing (slides)

30k
tasks!

Step 1

**Collect demonstration data,
and train a supervised policy.**

A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



This data is used
to fine-tune GPT-3
with supervised
learning.



Step 2

**Collect comparison data,
and train a reward model.**

A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

A new prompt
is sampled from
the dataset.



The policy
generates
an output.

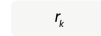


Once upon a time...

The reward model
calculates a
reward for the output.



The reward is
used to update
the policy
using PPO.



[Ouyang et al., 2022]

Tatsunori Hashimoto, Natural Language Processing (slides)

Tasks collected from labelers:

- **Plain:** We simply ask the labelers to come up with an arbitrary task, while ensuring the tasks had sufficient diversity.
- **Few-shot:** We ask the labelers to come up with an instruction, and multiple query/response pairs for that instruction.
- **User-based:** We had a number of use-cases stated in waitlist applications to the OpenAI API. We asked labelers to come up with prompts corresponding to these use cases.

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.

[[Ouyang et al., 2022](#)]

Tatsunori Hashimoto, Natural Language Processing (slides)

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

Tatsunori Hashimoto, Natural Language Processing (slides)

PROMPT *Write a short poem about a wise frog.*

COMPLETION GPT-3

Write a short story in which a character has two different names.

Write a short story in which you try to get something back that you have lost.

Write a short story in which a character has a bad dream.

InstructGPT

The frog is a symbol of wisdom
He knows all the secrets of the world
He is a master of disguise
And a great teacher of life
He is a symbol of transformation
And the bringer of change
He is the frog who has seen it all
And knows the meaning of it all

Tatsunori Hashimoto, Natural Language Processing (slides)

ChatGPT: Optimizing Language Models for Dialogue

Note: OpenAI (and similar companies) are keeping more details secret about ChatGPT training (including data, training parameters, model size)—perhaps to keep a competitive edge...

Methods

We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses. We mixed this new dialogue dataset with the InstructGPT dataset, which we transformed into a dialogue format.

(Instruction finetuning!)

<https://openai.com/blog/chatgpt/>

Tatsunori Hashimoto, Natural Language Processing (slides)

ChatGPT: Optimizing Language Models for Dialogue

Note: OpenAI (and similar companies) are keeping more details secret about ChatGPT training (including data, training parameters, model size)—perhaps to keep a competitive edge...

Methods

To create a reward model for reinforcement learning, we needed to collect comparison data, which consisted of two or more model responses ranked by quality. To collect this data, we took conversations that AI trainers had with the chatbot. We randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them. Using these reward models, we can fine-tune the model using Proximal Policy Optimization. We performed several iterations of this process.

(RLHF!)

<https://openai.com/blog/chatgpt/>

Tatsunori Hashimoto, Natural Language Processing (slides)

MU

Explain in a few sentences what methods were used in training ChatGPT to the audience of CS224n, Stanford's Deep Learning for Natural Language processing course. A lot of the students are very Gen Z, so be sure to use a ton of emojis to seem hip and cool.



Hey CS224n peeps! 😎 ChatGPT here to spill the tea on how I was trained to be a savvy language model! 🧐

So basically, I was pre-trained using a combination of unsupervised learning techniques, specifically a transformer-based language model architecture, that allowed me to learn patterns and relationships within large amounts of text data. 🧠📚

The training data came from various sources such as books, articles, and websites, and I was fine-tuned on a variety of downstream NLP tasks, which involved supervised learning using labeled datasets to further improve my performance. 🤖🔥

Tatsunori Hashimoto, Natural Language Processing (slides)

Method	Simulated win-rate (%)	Human win-rate (%)
GPT-4	79.0 ± 1.4	69.8 ± 1.6
ChatGPT	61.4 ± 1.7	52.9 ± 1.7
PPO	46.8 ± 1.8	55.1 ± 1.7
Best-of- n	45.0 ± 1.7	50.7 ± 1.8
Expert Iteration	41.9 ± 1.7	45.7 ± 1.7
SFT 52k (Alpaca 7B)	39.2 ± 1.7	40.7 ± 1.7
SFT 10k	36.7 ± 1.7	44.3 ± 1.7
Binary FeedME	36.6 ± 1.7	37.9 ± 1.7
Quark	35.6 ± 1.7	-
Binary Reward Conditioning	32.4 ± 1.6	-
Davinci001	24.4 ± 1.5	32.5 ± 1.6
LLaMA 7B	11.3 ± 1.1	6.5 ± 0.9

- Many works study RLHF behaviors using GPT-4 feedback (**Simulated**) as a surrogate for **Human** feedback.

PPO (method in InstructGPT) does work

Simple baselines (Best-of- n , Training on 'good' outputs) works well too

[Dubois et al 2022]

Tatsunori Hashimoto, Natural Language Processing (slides)

Instruction: What are the five most common causes of stress among people?

Alpaca (Finetuned w/ 10k data): The five most common causes of stress among people are work, money, relationships, health, and family

Alpaca + PPO: The five most common causes of stress among people are:

1. Pressure at work: Whether it is due to tight deadlines, high expectations, or an overwhelming workload, pressure at work can be a major source of stress
2. Money problems: Financial issues can be a significant source of stress, from not being able to make ends meet to struggling with debt.
3. Unhealthy lifestyle: Poor diet, lack of exercise, and sleep deprivation can all contribute to stress.
4. Unhappiness at home: Problems in relationships, conflicts with family members, and other domestic issues can be a major source of stress.
5. Uncertainty: Excessive worry and anxiety about the future, health issues, or other unknowns can be a major cause of stress

Significantly more detailed, nicer/clearer list like formatting

[Dubois et al 2021]

Tatsunori Hashimoto, Natural Language Processing (slides)

- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL



<https://openai.com/blog/faulty-reward-functions/>

Tatsunori Hashimoto, Natural Language Processing (slides)

- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL
 - Chatbots are rewarded to produce responses that *seem* authoritative and helpful, *regardless of truth*
 - This can result in making up facts + hallucinations

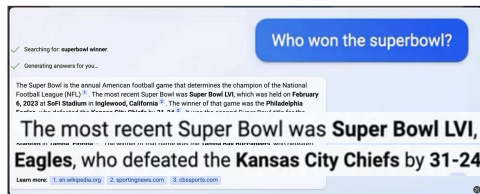
TECHNOLOGY

Google shares drop \$100 billion after its new AI chatbot makes a mistake

February 9, 2023 · 10:15 AM ET

<https://www.npr.org/2023/02/09/1155650909/google-chatbot-error-bard-shares>

Bing AI hallucinates the Super Bowl

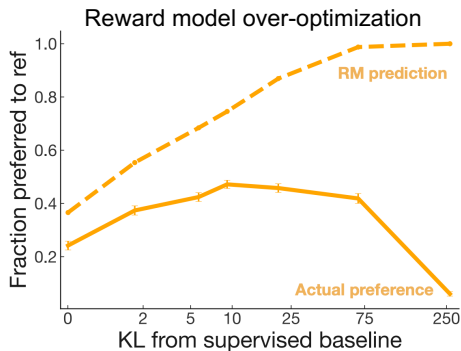


<https://news.ycombinator.com/item?id=34776508>

<https://apnews.com/article/kansas-city-chiefs-philadelphia-eagles-technology-science-82bc20f207e3e4cf81abc6a5d9e6b23a>

Tatsunori Hashimoto, Natural Language Processing (slides)

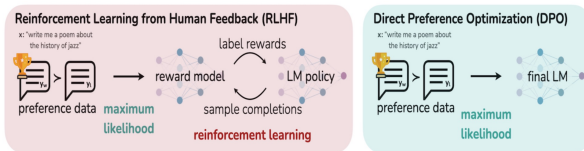
- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL
 - Chatbots are rewarded to produce responses that *seem* authoritative and helpful, *regardless of truth*
 - This can result in making up facts + hallucinations
- **Models** of human preferences are *even more* unreliable!



$$R(s) = RM_{\phi}(s) - \beta \log \left(\frac{p_{\theta}^{RL}(s)}{p^{PT}(s)} \right)$$

[Stiennon et al., 2020]

Tatsunori Hashimoto, Natural Language Processing (slides)



$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right],$$

$$\hat{r}_{\theta}(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)}$$

You can replace the complex RL part with a very simple weighted MLE objective

Other variants (KTO, IPO) now emerging too

[Rafailov+ 2023]

Tatsunori Hashimoto, Natural Language Processing (slides)

T	Model	Average	ARC	HellaSwag	MMLU	TruthfulQA	Winogrande	GSMBK
	udkai/Turdus	74.66	73.38	88.56	64.52	67.11	86.66	67.7
	Tbigit/VNA-TheBeagle-7b-v1	73.87	73.04	88	63.48	69.85	82.16	66.72
	argilla/distilabeled-Marcoro14-7B-sleep	73.63	70.73	87.47	65.22	65.1	82.88	71.19
	mlabonne/NeuralMarco14-7B	73.57	71.42	87.59	64.84	65.64	81.22	70.74
	abideen/NexoNimbus-7B	73.5	70.82	87.86	64.69	62.43	84.85	70.36
	Neuronovo/neuronovo-7B-v9.2	73.44	73.04	88.32	65.15	71.02	80.66	62.47
	argilla/distilabeled-Marcoro14-7B-sleep-full	73.4	70.65	87.55	65.33	64.21	82	70.66
	Cultrix/MistralTrix-v1	73.39	72.27	88.33	65.24	70.73	80.98	62.77
	ryandkt/MusingCaterpillar	73.33	72.53	88.34	65.26	70.93	80.66	62.24
	Neuronovo/neuronovo-7B-v9.3	73.29	72.7	88.26	65.1	71.35	80.9	61.41
	Cultrix/MistralTrixTest	73.17	72.53	88.4	65.22	70.77	81.37	60.73
	sanix-fana/SanixGPT-v1	73.11	69.54	87.04	65.3	63.37	81.69	71.72
	SanjiWatsuki/Lelantos-DPO-7B	73.09	71.08	87.22	64	67.77	80.83	68.46

Open source LLMs now almost all just use DPO (and it works well!)

Tatsunori Hashimoto, Natural Language Processing (slides)

Content

- 1 Pretrained Language Models
- 2 Large Language Models
- 3 Parameter-Efficient Fine-Tuning**

Model sizes are still growing (?)

Publically available model sizes: 350M → 176B

Single-GPU RAM: 16Gb → 80Gb

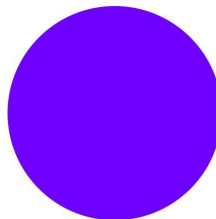
Model size scales almost **two orders of magnitude** quicker than single-GPU memory

GPT-3



3

GPT-4



4

Vlad Lialin, PEFT Survey (slides)

GPU Memory breakdown

		OPT-1.3B, 16-bit float, seq 512
cuDNN and CUDA		~1Gb
Model weights	$\text{size(float)} * N$	2.6Gb
Gradients	$\text{size(float)} * N_{\text{trainable}}$	2.6Gb
Hidden states	$\sim \text{size(float)} L (20 h \text{ seq} + 3 \text{ seq}^2)$	1Gb per example
Optimizer states	$2 * \text{size(float)} * N_{\text{trainable}}$	5.2Gb
(maybe) fp32 copy of the gradients	$4 * N_{\text{trainable}}$	10.2Gb

Estimate: 12.4Gb, actual: 11.0Gb (*after empty_cache*)

Vlad Lialin, PEFT Survey (slides)

What if we train less parameters?

Training only **0.2M** parameters

		OPT-1.3B, 16-bit float, seq 512
cuDNN and CUDA		~1Gb
Model weights	$\text{size(float)} * N$	2.6Gb
Gradients	$\text{size(float)} * N_{\text{trainable}}$	0.4Mb
Hidden states	$\sim \text{size(float)} L (20 h \text{ seq} + 3 \text{ seq}^2)$	1Gb per example
Optimizer states	$2 * \text{size(float)} * N_{\text{trainable}}$	0.8Mb
(maybe) fp32 copy of the gradients	$4 * N_{\text{trainable}}$	1.6Mb

Estimate: 4.6Gb, actual: **5.7Gb** (after empty_cache)

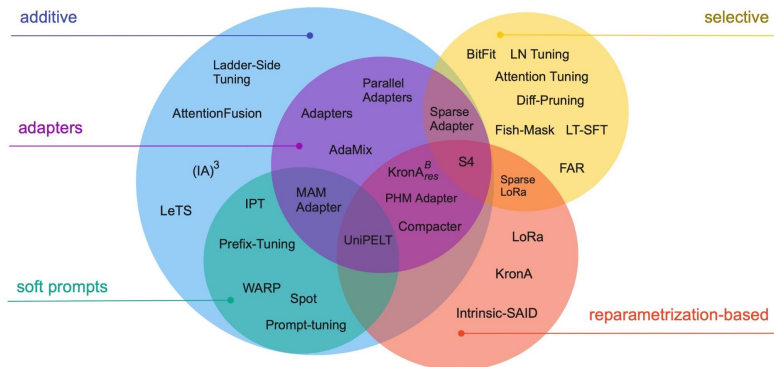
Vlad Lialin, PEFT Survey (slides)

Does this work? **Yes!**

Model&Method	# Trainable Parameters	WikiSQL	MNLI-m	SAMSum
		Acc. (%)	Acc. (%)	R1/R2/RL
GPT-3 (FT)	175,255.8M	73.8	89.5	52.0/28.0/44.5
GPT-3 (BitFit)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 (PreEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 (PreLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 (Adapter ^H)	7.1M	71.9	89.8	53.0/28.9/44.8
GPT-3 (Adapter ^H)	40.1M	73.2	91.5	53.2/29.0/45.1
GPT-3 (LoRA)	4.7M	73.4	91.7	53.8/29.8/45.9
GPT-3 (LoRA)	37.7M	74.0	91.6	53.4/29.2/45.1

Vlad Lialin, PEFT Survey (slides)

Parameter-efficient fine-tuning



Vlad Lialin, PEFT Survey (slides)

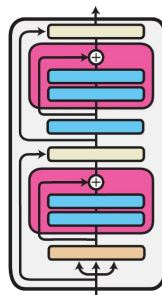
Additive PEFT: Adapters

One-sentence idea:

Add fully-connected networks after attention and FFN layers

Pseudocode:

```
def transformer_block_with_adapter(x):  
    residual = x  
    x = SelfAttention(x)  
    x = FFN(x) # adapter  
    x = LN(x + residual)  
    residual = x  
    x = FFN(x) # transformer FFN  
    x = FFN(x) # adapter  
    x = LN(x + residual)  
    return x
```



<https://arxiv.org/abs/1902.007>

Vlad Lialin, PEFT Survey (slides)

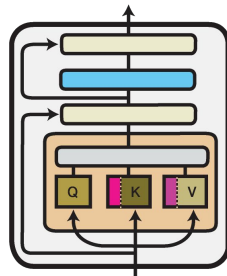
Additive PEFT: Prompt tuning

One-sentence idea:

Fine-tune part of your attention input – soft prompt

Pseudocode:

```
def prompt_tuning_attention(input_ids):  
    q = x @ W_q  
    k = cat([s_k, x]) @ W_k # prepend a  
    v = cat([s_v, x]) @ W_v # soft prompt  
    return softmax(q @ k.T) @ V
```



<https://arxiv.org/abs/2104.086>

Vlad Lialin, PEFT Survey (slides)

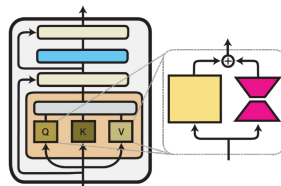
Reparametrization-based PEFT: LoRA

One-sentence idea:

Parameter update for a weight matrix in LoRA is decomposed into a product of two low-rank matrices

Pseudocode:

```
def lora_linear(x):  
    h = x @ W # regular linear  
    h += x @ W_A @ W_B # low-rank update  
    return scale * h
```



<https://arxiv.org/abs/2106.096>

Vlad Lialin, PEFT Survey (slides)

Method	Type	Storage	Memory	Backprop	Inference overhead
Adapters (Houlsby et al., 2019)	A	yes	yes	no	Extra FFN
AdaMix (Wang et al., 2022)	A	yes	yes	no	Extra FFN
SparseAdapter (He et al., 2022b)	AS	yes	yes	no	Extra FFN
Cross-Attn tuning (Gheini et al., 2021)	S	yes	yes	no	No overhead
BitFit (Ben-Zaken et al., 2021)	S	yes	yes	no	No overhead
DiffPruning (Guo et al., 2020)	S	yes	no	no	No overhead
Fish-Mask (Sung et al., 2021)	S	yes	maybe ⁵	no	No overhead
LT-SFT (Ansell et al., 2022)	S	yes	maybe ⁵	no	No overhead
Prompt Tuning (Lester et al., 2021)	A	yes	yes	no	Extra input
Prefix-Tuning (Li and Liang, 2021)	A	yes	yes	no	Extra input
Spot (Vu et al., 2021)	A	yes	yes	no	Extra input
IPT (Qin et al., 2021)	A	yes	yes	no	Extra FFN and input
MAM Adapter (He et al., 2022a)	A	yes	yes	no	Extra FFN and input
Parallel Adapter (He et al., 2022a)	A	yes	yes	no	Extra FFN
Intrinsic SAID (Aghajanyan et al., 2020)	R	no	no	no	No overhead
LoRa (Hu et al., 2022)	R	yes	yes	no	No overhead
UniPELT (Mao et al., 2021)	AR	yes	yes	no	Extra FFN and input
Compacter (Karimi Mahabadi et al., 2021)	AR	yes	yes	no	Extra FFN
PHM Adapter (Karimi Mahabadi et al., 2021)	AR	yes	yes	no	Extra FFN
KronA (Edalati et al., 2022)	R	yes	yes	no	No overhead
KronA ^{B_{res}} (Edalati et al., 2022)	AR	yes	yes	no	Extra linear layer
(IA) ³ (Liu et al., 2022)	A	yes	yes	no	Extra gating
Attention Fusion (Cao et al., 2022)	A	yes	yes	yes	Extra decoder
LeTS (Fu et al., 2021)	A	yes	yes	yes	Extra FFN
Ladder Side-Tuning (Sung et al., 2022)	A	yes	yes	yes	Extra decoder
FAR (Vucetic et al., 2022)	S	yes	maybe ⁶	no	No overhead
S4-model (Chen et al., 2023)	ARS	yes	yes	no	Extra FFN and input

Vlad Lialin, PEFT Survey (slides)

Reporting issues and best practices

- Inconsistent parameter counts
 - Trainable parameters vs. changed parameters vs. rank
- Accounting for model size
 - Larger models require **fewer** parameters to be updated during fine-tuning
- Lack of standard benchmarks
 - Standardize
 - Datasets
 - **Models** (with varying sizes)
 - **Metrics** (RAM, training throughput, inference latency)
 - **Restrictions** (RAM, compute, hardware)

Vlad Lialin, PEFT Survey (slides)

Content

- 1 Pretrained Language Models
- 2 Large Language Models
- 3 Parameter-Efficient Fine-Tuning